

It is executed through the operating system and existing software layers. This leads to challenges such as overheads in scheduling parallel tasks on these units, maintenance of data dependencies (which requires programmers to account for and maintain the status of memory contents when switching between different compute units such as CPUs and GPUs), workload partitioning (choosing an appropriate compute unit for a given workload), and reaching higher levels of energy efficiency and power saving.

Heterogeneous systems are still evolving and various developments are happening in areas like the operating system, virtual machine (ISA), tool chain and runtime libraries. Open source compilers and tools like GCC and LLVM are supporting the polyhedral model for heterogeneous architectures. There is a separate community driving polyhedral compilation research. Figure 6 provides an overview of the ecosystem influencing the evolution of HSA.

NVIDIA's CUDA alone could not cater to the evolving aspects of heterogeneous systems. Newer open platforms, standards, architectures like HSA Foundation, OpenCL, OpenACC, IBM's Liquid Metal, and OpenHPC have emerged to provide the software stack for heterogeneous systems. Various open source projects are being run as part of the HSA Foundation, which one can leverage and contribute back to.

A distributed computing platform like Apache Spark and others have contributed a lot to the evolution of heterogeneous systems. As a result, platforms like SparkCL and HeteroSpark have evolved, which combine the scale-out power of distributed computing and the scale-up power of heterogeneous computing to create a very powerful HPC platform. Efforts are on to use the advancements made in NVM (non-volatile memory) technology in heterogeneous systems. OpenARC (Open Accelerator Research Compiler) is one such effort.

In short, it is clear that the mainstream hardware is becoming permanently parallel, heterogeneous and distributed. These changes are here to stay, and will

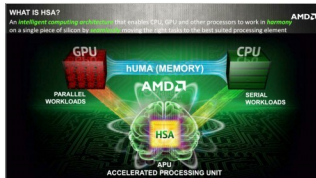


Figure 5: AMD's heterogeneous high performance computing architecture
(Source: http://www.overclock3d.net/news/software/amd_team_up_with_microsoft_for_hsa_c1/)



Figure 6: The HSA ecosystem

disrupt the way we have to write performance-intensive code on mainstream architecture. This evolution of heterogeneous architecture clearly highlights the increasing demand for compute-efficient architectures. There are various proprietary heterogeneous architectures that address today's challenges in their own way. The increasing demand for such architectures shows that the heterogeneous system architecture is inevitable and is the way forward. We will discuss the components of heterogeneous system architectures in depth in the forthcoming articles in this series. **END**

References

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- [6] A list of open source projects is available at the Foundation's GitHub repository (<http://www.hsafoundation.com/hsaf-open-source-developer-program/>)
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